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LICENCIATURA

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ASIGNATURA

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TEMA

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Architecture and Attacks in Software-Defined Networking (SDN) and its Security Using Machine Learning

The paper explores Software-Defined Networking, a paradigm that decouples the control plane from the data plane across a three layer layout (Application, Control, and Infrastructure) tied together through Northbound and Southbound APIs (such as OpenFlow). While this separation brings flexibility, programmability, and centralized management, it also introduces systemic vulnerabilities across every layer and interface. The study highlights Distributed Denial of Service (DDoS) assaults as a critical hazard, noting how malicious request floods can exhaust the central controller's memory, bandwidth, and processing power, crippling overall network availability and administration.

Fiber-Wireless Broadband Access: Convergence: Past, Present, and Future

This presentation details FiWi systems, which merge optical fiber backbones with wireless access points. It outlines a tripartite structure comprising central offices, optical access through PON, and wireless edges alongside the migration toward FTTx and convergence frameworks like Radio over Fiber (RoF). Although these configurations deliver scalable bandwidth, quality of service, and flexible deployment, they struggle with optical network unit (ONU) placement, interference management, and energy conservation (Green FiWi), while targeting smart grid applications.

Architecture of Ad-Hoc Networks (Polibits, IPN)

The review examines dynamic, infrastructureless peer-to-peer networks where every participating device functions simultaneously as an end host and a packet router. Grouped into categories such as MANETs, Bluetooth, wireless sensor networks (WSNs), and WSANs, their technical stack spans physical layers running on ISM bands via IEEE 802.11, MAC layers utilizing CSMA/CA, and network layers driven by proactive (OLSR, TBRPF) or reactive (AODV, DSR) routing protocols, with real world deployments found in consumer electronics and military systems like FBCB2.

IoT Network Architecture and Security Fundamentals

With over 16.7 million connected gadgets operating globally, safeguarding Internet of Things environments has turned into a major engineering hurdle. The overview discusses expanding attack surfaces where ordinary appliances act as hacker entry points, the catastrophic ripple effects of smart-city breaches such as power grid

shutdowns leading to physical damage and the shared duty of manufacturers and developers to bake security into systems from day one using zero-trust and DevSecOps frameworks.

Information Centric Networks (ICN)

Pointing out that legacy Internet architecture was built for point-to-point host communication rather than modern content delivery, this approach champions Information-Centric Networking as a clean-slate fix for the Future Internet. ICN shifts the focus from host addresses to content retrieval names, natively supporting mobility, multihoming, and enhanced security by trading channel based safeguards for content level cryptographic signatures verified via public keys, though commercial-scale rollout still requires deep testing.

A Survey of Information-Centric Networking Research (IEEE)

This deep dive into ICN mechanics focuses on hierarchical and flat naming schemes, in-network caching, receiver-driven routing, and publish-subscribe models. Evaluating architectures like NDN, NetInf, and PURSUIT, the study establishes ICN as a solid blueprint for future web infrastructure, albeit one that must first conquer hurdles in routing scalability, cache consistency, and access security.

Trust in 5G and Future Networks

This presentation traces the evolution from rigid perimeter defenses in older cellular generations to continuous, distributed trust models in 5G and beyond. Key architectural pillars include network slicing for isolated sectors like healthcare and industry, mobile edge computing that widens attack vectors, and automated orchestration managed by explainable AI. Relying on a Zero Trust Architecture framework featuring continuous verification, slice isolation, and edge monitoring, these systems secure high-stakes domains like autonomous transit and smart manufacturing.